

实验四：PIN 二极管特性仿真

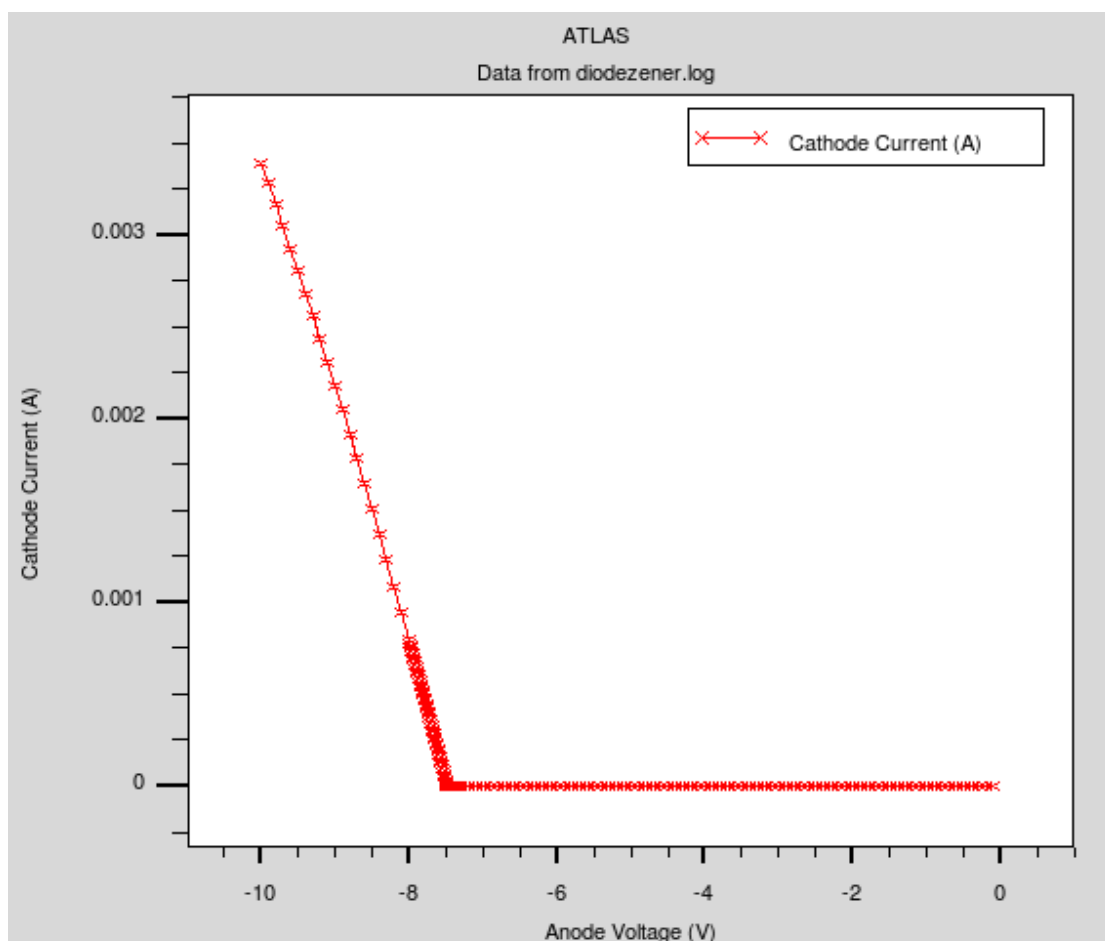
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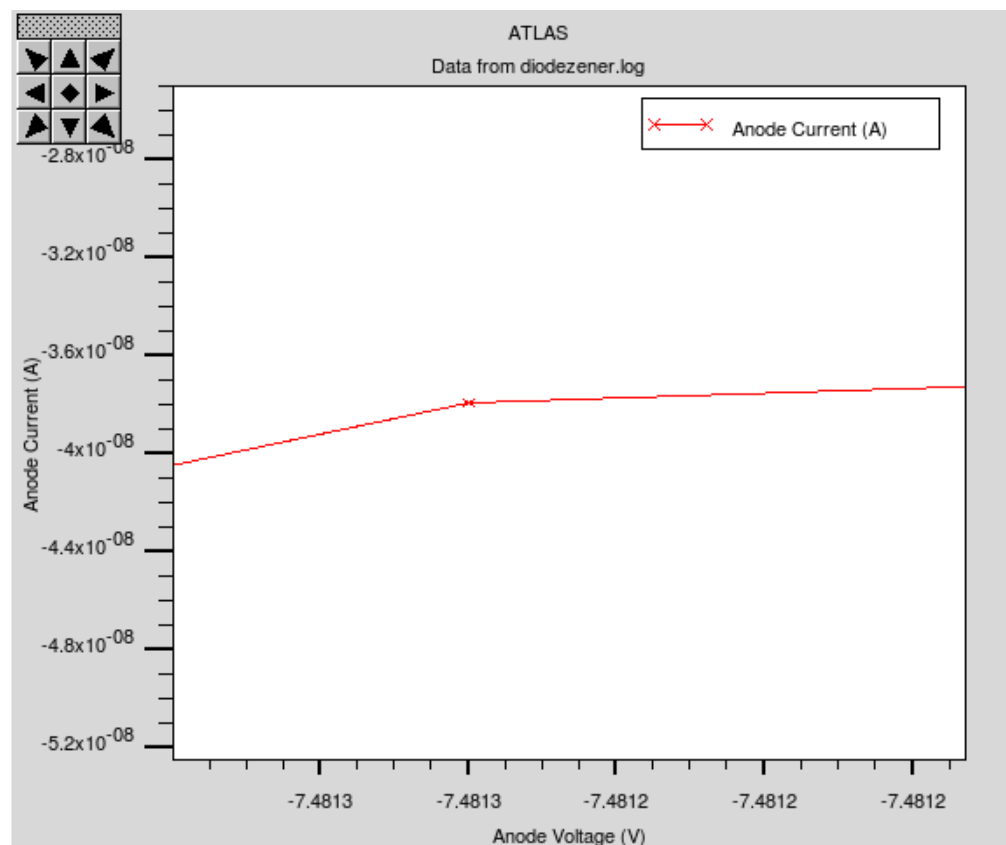
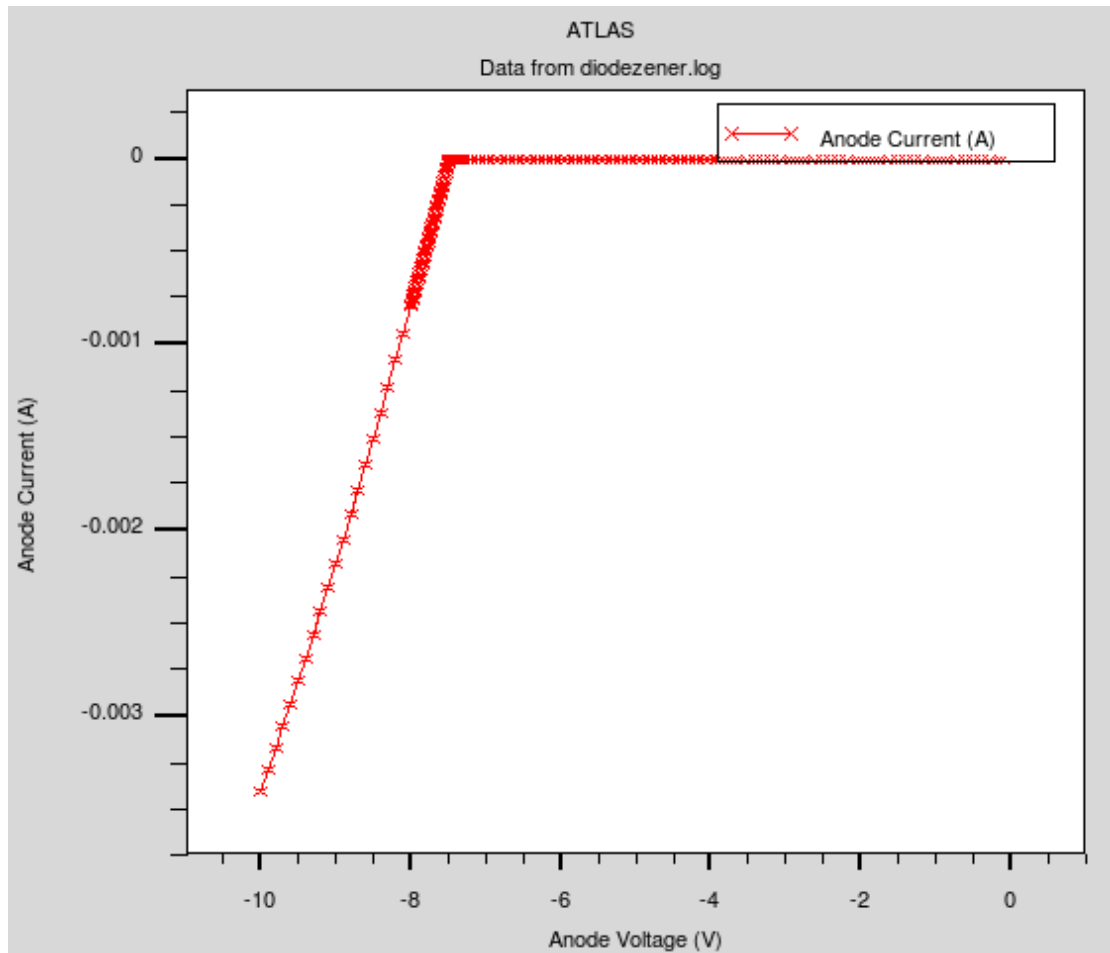
时间： 2025/11/12 ；

- **Requirements:** Simulate the reverse breakdown voltage of the PIN diode when $d=0, 0.2$, and 0.4 micrometers, and plot the relationship between the breakdown voltage of the PIN diode and d .

模拟当 $d=0$ 、 0.2 和 $0.4 \mu\text{m}$ 时的 PIN 二极管的反向击穿电压，并绘制 PIN 二极管的击穿电压与 d 之间的关系图。

1 $d=0\mu\text{m}$





从图中可知，击穿电压： **$U=-7.4813V$**

运行代码：

```
go atlas
```

```
mesh
```

```
x.mesh location=0.0 spac=0.1
```

```
x.mesh location=1.0 spac=0.1
```

```
y.mesh location=0    spac=0.1
```

```
y.mesh location=0.9    spac=0.001
```

```
y.mesh location=1.8 spac=0.1
```

```
region num = 1 y.min = 0 y.max = 0.9 material = silicon
```

```
#region num = 2 y.min = 0.9 y.max = 2.9 material = silicon
```

```
region num = 3 y.min = 0.9 y.max = 1.8 material = silicon
```

```
electrode name=anode top
```

```
electrode name=cathode bottom
```

```
doping region = 1 p.type uniform conc = 1e18
```

```
doping region = 2 n.type uniform conc = 5e10
```

```
doping region = 3 n.type uniform conc = 1e18
```

output con.band val.band QFN QFP

models bipolar bbt.std print

impact selb

method newton trap maxtrap=30 \

climit=1e-4

solve init

log outfile=diodezener.log

solve vanode=-0.1 vstep=-0.1 vfinal=-7.2 name=anode

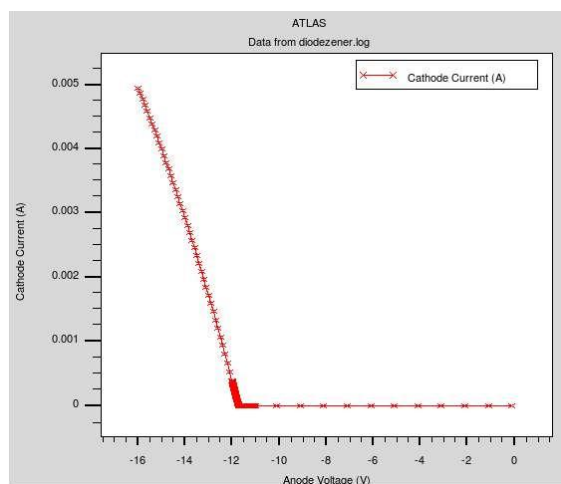
solve vanode=-7.3 vstep=-0.01 vfinal=-8.0 name=anode

solve vanode=-8.0 vstep=-0.1 vfinal=-10 name=anode

tonyplot diodezener.log

quit

2 d=0.2 μ m



从图中可知，击穿电压： **$U=-11.729V$**

运行代码：

```
go atlas
```

```
mesh
```

```
x.mesh location=0.0 spac=0.1
```

```
x.mesh location=1.0 spac=0.1
```

```
y.mesh location=0    spac=0.1
```

```
y.mesh location=0.9    spac=0.005
```

```
y.mesh location=1.1    spac=0.005
```

```
y.mesh location=2.0 spac=0.1
```

```
region num = 1 y.min = 0 y.max = 0.9 material = silicon
```

```
region num = 2 y.min = 0.9 y.max = 1.1 material = silicon
```

```
region num = 3 y.min = 1.1 y.max = 2.0 material = silicon
```

```
electrode name=anode top
```

```
electrode name=cathode bottom
```

```
doping region = 1 p.type uniform conc = 1e18
```

```
doping region = 2 n.type uniform conc = 5e10
```

```
doping region = 3 n.type uniform conc = 1e18
```

output con.band val.band QFN QFP

models bipolar bbt.std print

impact selb

method newton trap maxtrap=30 \

climit=1e-4

solve init

log outfile=diodezener.log

solve vanode=-0.1 vstep=-1 vfinal=-10 name=anode

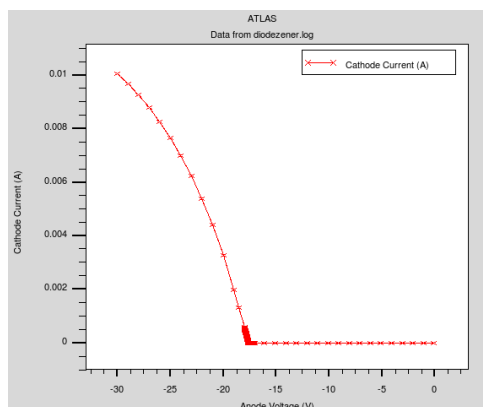
solve vanode=-11 vstep=-0.001 vfinal=-12 name=anode

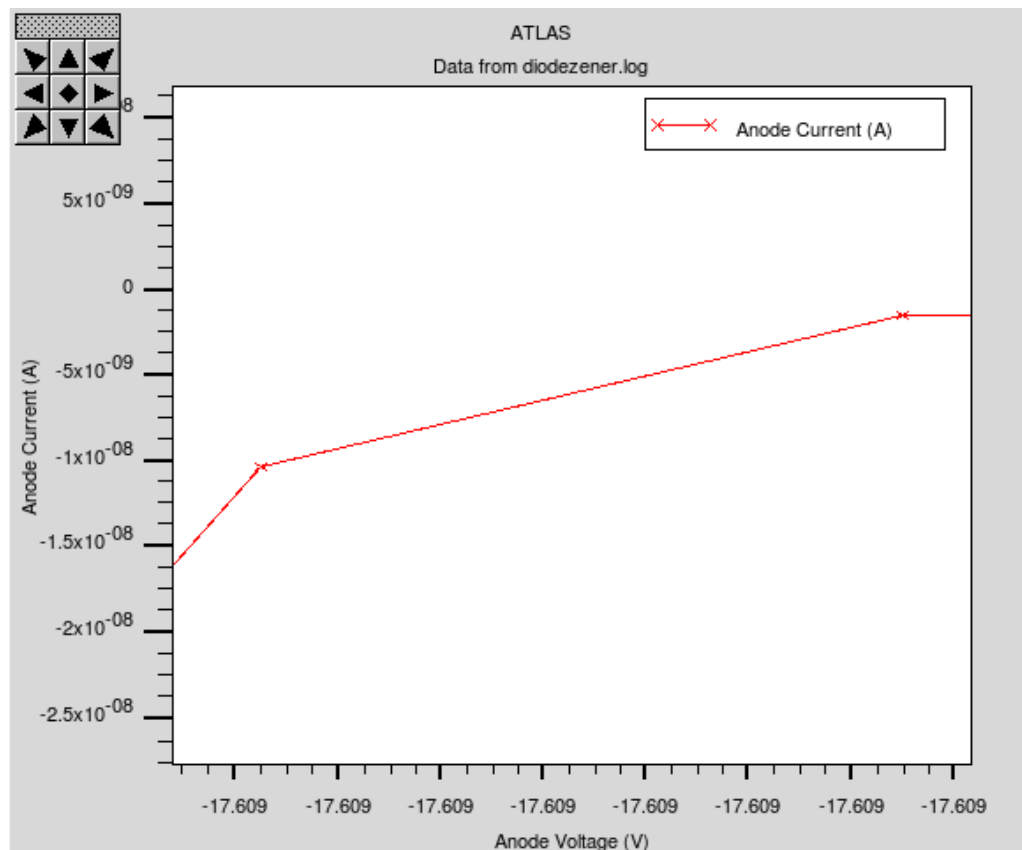
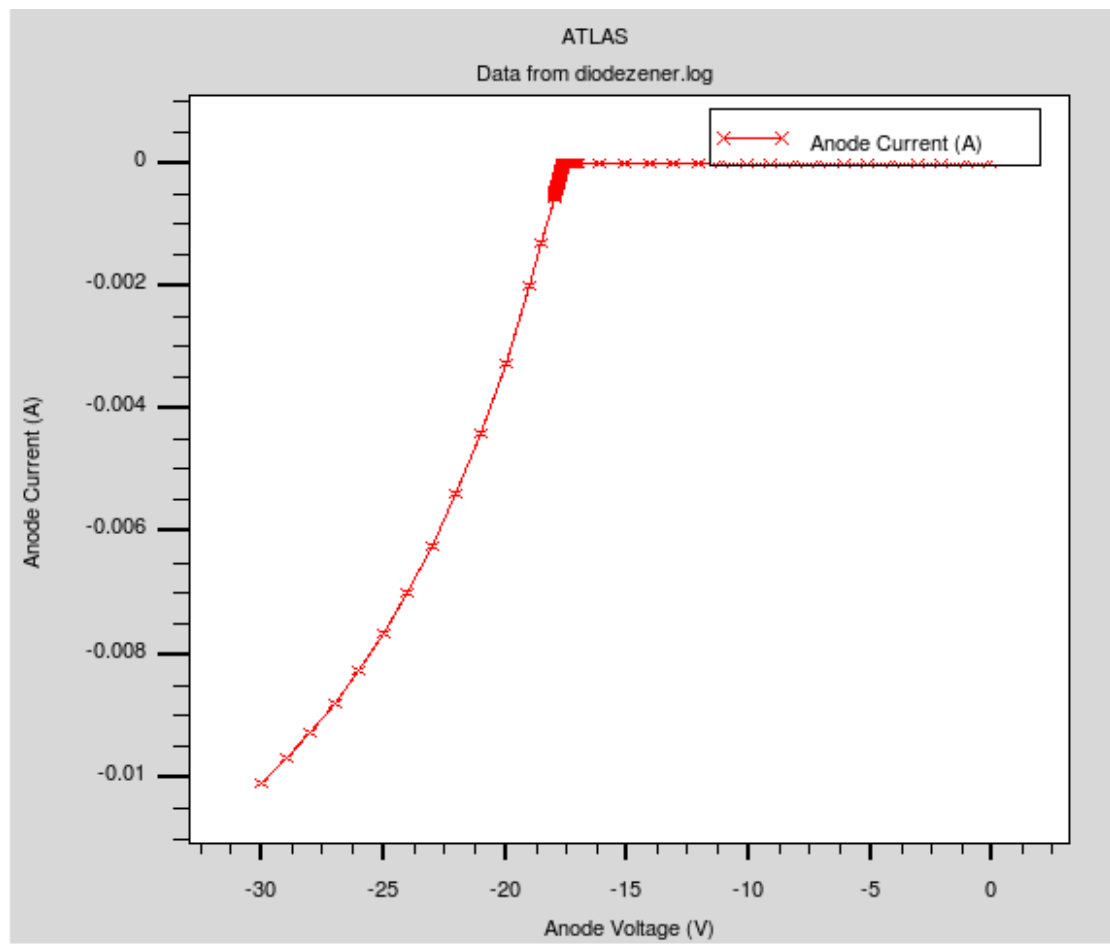
solve vanode=-12 vstep=-0.1 vfinal=-16 name=anode

tonyplot diodezener.log

quit

3 d=0.4 μ m





从图中可知，击穿电压： **$U=-17.609V$**

运行代码：

```
go atlas
```

```
mesh
```

```
x.mesh location=0.0 spac=0.1
```

```
x.mesh location=1.0 spac=0.1
```

```
y.mesh location=0    spac=0.1
```

```
y.mesh location=0.9    spac=0.005
```

```
y.mesh location=1.3    spac=0.005
```

```
y.mesh location=2.2 spac=0.1
```

```
region num = 1 y.min = 0 y.max = 0.9 material = silicon
```

```
region num = 2 y.min = 0.9 y.max = 1.3 material = silicon
```

```
region num = 3 y.min = 1.3 y.max = 2.2 material = silicon
```

```
electrode name=anode top
```

```
electrode name=cathode bottom
```

```
doping region = 1 p.type uniform conc = 1e18
```

```
doping region = 2 n.type uniform conc = 5e10
```

```
doping region = 3 n.type uniform conc = 1e18
```

```
output con.band val.band QFN QFP
```

```
models bipolar bbt.std print
```

```
impact selb
```

```
method newton trap maxtrap=30 \
```

```
climit=1e-4
```

```
solve init
```

```
log outfile=diodezener.log
```

```
solve vanode=-0.1 vstep=-1 vfinal=-17 name=anode
```

```
solve vanode=-17 vstep=-0.001 vfinal=-18 name=anode
```

```
solve vanode=-18 vstep=-1 vfinal=-30 name=anode
```

```
tonyplot diodezener.log
```

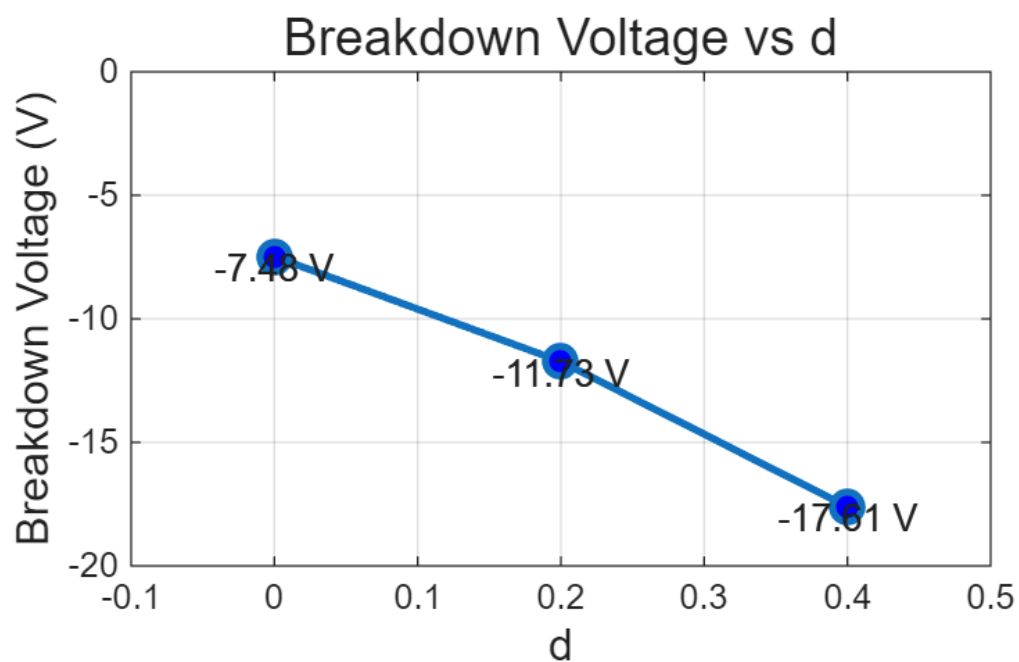
```
quit
```

4 PIN 二极管的击穿电压与 d 之间的关系

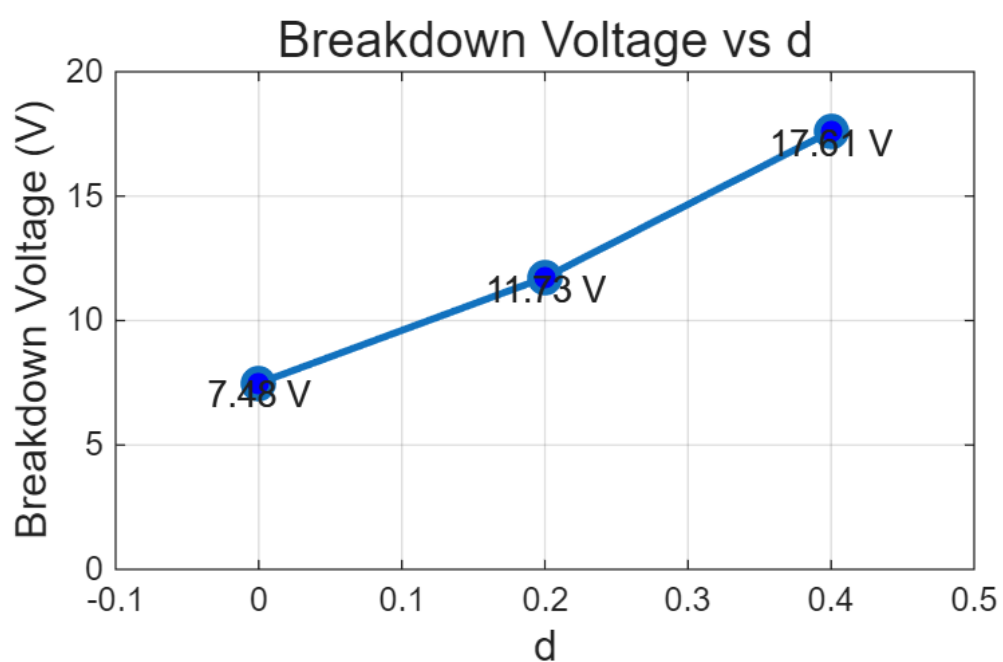
实验数据记录如下：

d	$V_{\text{breakdown}}$	
0	-7.4813	
0.2	-11.729	
0.4	-17.609	

绘制如图：



击穿电压取正值时：



因为 PIN 二极管的 I 层几乎无掺杂，电场在其中近似均匀分布，

击穿时满足 $E_c = V_{br}/d$,

所以击穿电压 $V_{br} = E_c \cdot d$ ，与 I 层宽度 d 近似成线性关系。